

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0739

Aspen 138 kV
February 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0739. The Interconnection Customer proposes a 69.7 MW solar photovoltaic facility interconnecting at the Aspen 138 kV substation in the Virginia area. The request was received on February 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 227 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,013,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
New London-Barksdale 138 KV	Carson Ice Cogen-Alderwood 138 KV	155.3	150	5.3
Coffeen N-Harrisonburg 138 KV	Crandall-Dayton 345 KV	114.5	350	5.3
Sta 54-Westchase 138 KV	Crandall-Dayton 345 KV	114.0	369	5.3
Westchase-Coffeen N 138 KV	Crandall-Dayton 345 KV line	112.8	355	5.3
Redland Substation-Verona 138 KV	Crandall-Dayton 345 KV	108.8	238	7.2
Greenbank-Verona 138 KV	Crandall-Dayton 345 KV	107.2	234	7.2
La Joie-Fairfield 138 KV line	Crandall-Aspen 345 KV	104.1	202	11.1
Alderwood-Carson Ice Cogen 138 KV	Barksdale-New London 138 KV L	102.0	271	5.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
New London-Barksdale 138 KV	Rebuild line with higher-capacity conductor	\$15,083,000	16% (\$2,484,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Coffeen N-Harrisonburg 138 KV	Rebuild line with higher-capacity conductor	\$1,357,000	5% (\$67,000)
Sta 54-Westchase 138 KV	Rebuild line with higher-capacity conductor	\$4,504,000	5% (\$223,000)
Westchase-Coffeen N 138 KV line	Rebuild line with higher-capacity conductor	\$5,767,000	5% (\$285,000)
Verona-Redland 138 KV	Reconductor line	\$1,618,000	7% (\$114,000)
Greenbank-Verona 138 KV	Reconductor line	\$1,153,000	7% (\$81,000)
La Joie-Fairfield 138 KV	Reconductor line	\$14,877,000	10% (\$1,482,000)
Alderwood-Carson Ice Cogen 138 KV	Reconductor line	\$1,727,000	16% (\$277,000)
	Total allocated to project		\$5,013,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.